

AI-Based Solution for Analysing, Benchmarking and Quality Monitoring of Cybersecurity Audit Reports

USER MANUAL

Objective 1: Standardisation of Cybersecurity Audit Report Format

Software component: Report-to-JSON Document Standardisation Pipeline

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1. Introduction

1.1 Purpose of this Manual

This User Manual provides step-by-step guidance for operating the Standardisation module of the **AI-Based Solution for Analysing, Benchmarking and Quality Monitoring of Cybersecurity Audit Reports and Performance Monitoring of Auditing Organisations** (hereafter “the Platform”). It explains the key functions, the inputs required from the user, the outputs the Platform produces, and the basic troubleshooting actions available to the user.

1.2 Scope

This manual covers Objective 1: Standardisation of Cybersecurity Audit Report Format. It describes how authorised users upload cybersecurity audit reports, how the Platform converts unstructured or semi-structured reports into the standard, machine-readable and source-traceable format, and how reviewers verify and approve the structured output. The dashboard, benchmark assessment, benchmark definition and agency performance modules (Objectives 2 to 5) are documented in separate manuals.

The initial release supports four audit domains: **Vulnerability Assessment and Penetration Testing (VAPT)**, **Network Infrastructure Audit**, **Mobile Application Security Audit** and **Application Security Testing (including Web Application Security Testing)**.

1.3 Intended Audience

User Group	How they use this manual
Analysts	Upload reports, monitor processing, correct submission details (Sections 4 and 5).
Reviewers	Verify AI-generated structured output against the source report; approve, edit, reject or override (Sections 6 to 8).
Decision-makers	View approved structured reports and exports; no editing rights (Sections 3, 6.7 and 9).
Administrators	Manage users, the report-structure update module, logs and backups (Sections 10 and 11).

1.4 Human Decision Authority

IMPORTANT: The Platform is an **AI-assisted decision-support system**. It does not renew, reject, suspend, blacklist or penalise any auditing agency, and it does not replace human auditors or reviewers. Every AI-generated field is a proposal that an authorised reviewer must verify. Final decisions remain only with authorised CERT-In / MeitY officials.

1.5 Conventions Used

Convention	Meaning
Bold text	Name of a screen, menu, button, field or status shown in the Platform.
NOTE	Additional information that helps you complete a task.
CAUTION / IMPORTANT	An action that may cause data loss, rejection of a file or a security issue.

Convention	Meaning
Step lists (1, 2, 3 ...)	Actions to be performed in the order given.
[Screenshot placeholder]	Position of an interface screenshot to be inserted from the released build.

2. System Overview

2.1 What the Platform Does

Audit reports from different agencies vary widely in structure, terminology, severity labels, evidence presentation and depth. The Platform removes this variation by (a) defining one standard audit report template and schema and (b) converting each submitted report into that schema using an offline, hybrid pipeline of rule-based parsing and AI/ML/NLP models. Every extracted item remains linked to its location in the original report.

The Platform operates entirely inside the approved **air-gapped environment**. It uses no internet connection, no cloud service and no online AI model; all models, databases, logs and artefacts are hosted locally.

2.2 Processing Pipeline

Each uploaded report passes through the twelve stages shown in Figure 1. Stages 1 to 11 are automated and monitored; stage 12 is reached only after a reviewer approves the output.

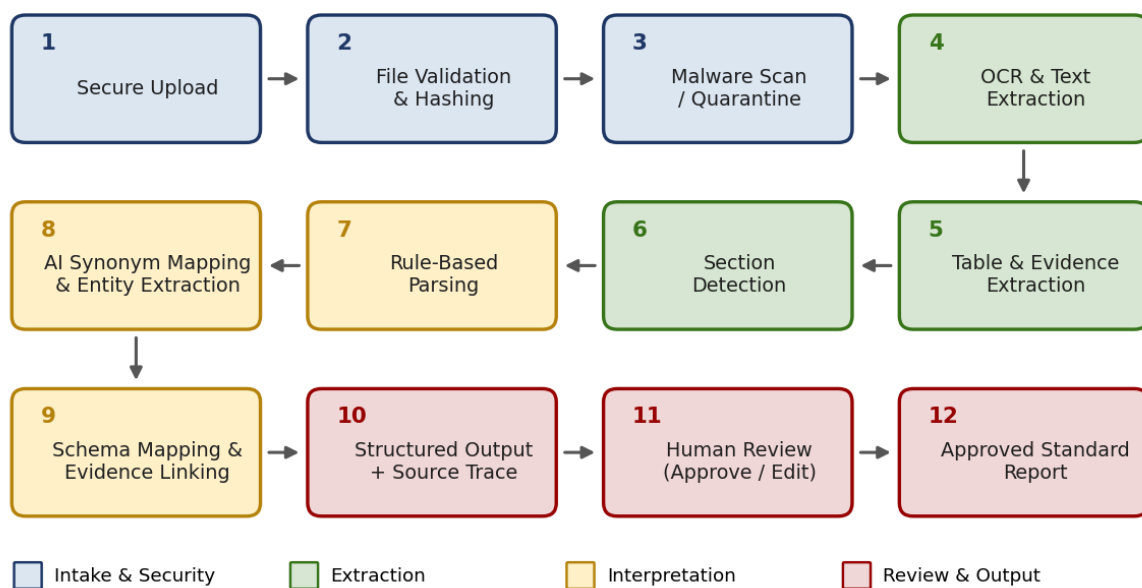


Figure 1: Report conversion pipeline

2.3 Supported Input Formats

Input type	How the Platform handles it
PDF (text-based)	Direct text, table and image extraction.
PDF (scanned) and image files (PNG, JPEG)	OCR is applied; screenshots are stored as evidence items.
DOCX and TXT	Text, headings, tables and embedded images are extracted.

Input type	How the Platform handles it
XLSX annexures	Each worksheet is read as a table and attached to the related finding or annexure.
Logs and tool outputs	Stored as evidence and linked to the finding they support.

NOTE: The maximum file size, page count and the exact list of accepted extensions are set by the Administrator and are displayed on the **Upload** screen.

2.4 User Roles and Permissions

Role	Upload	Review / Edit	Approve	View	Administer
Analyst	Yes	Own uploads only (metadata)	No	Assigned reports	No
Reviewer	No	Yes	Yes	Assigned reports	No
Decision-maker	No	No	No	Approved reports	No
Administrator	Yes	No (content)	Update module approvals	All	Yes

2.5 Access Requirements

- An authorised user account created by the Administrator, with a registered MFA device.
- A workstation within the approved secure network, using the browser specified in the deployment guide.
- Reports to be processed must be legally cleared for upload; anonymised real data may be processed only under written approval.

3. Getting Started

3.1 Logging In

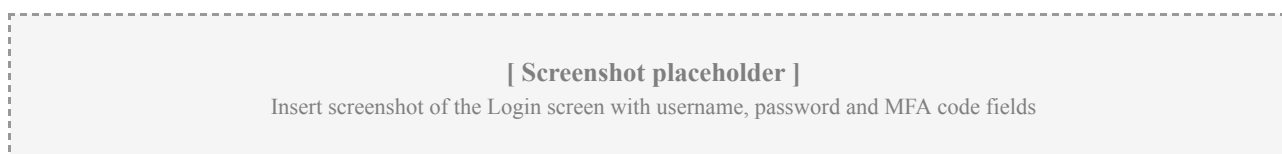


Figure 2: Login and MFA screen (indicative)

1. Open the Platform address given by your Administrator in the approved browser.
2. Enter your **Username** and **Password** and select **Sign In**.
3. When prompted, enter the one-time code generated by your registered MFA device and select **Verify**.
4. On success, the **Home** screen opens showing your role, assigned reports and recent activity.

CAUTION: Never share credentials or MFA codes. Repeated failed sign-ins lock the account; only the Administrator can unlock it. All sign-ins are written to the access log.

3.2 Navigating the Platform

The left-hand menu is role-based; you see only the options permitted to you.

Menu option	Purpose
Home	Summary of your queue, recent uploads and notifications.
Upload Report	Submit a report and its annexures (Section 4).
Processing Queue	Track status of every report you may access (Section 5).
Review Workspace	Verify, correct and approve structured output (Section 6).
Standard Template	Download the template, field guide and schema (Section 7).
Exports	Generate approved structured outputs (Section 9).
Administration	Users, update module, logs, backup (Sections 10 and 11). Administrators only.
Help	This manual, quick-reference card and support contact details.

3.3 Session Security

Your session ends automatically after a period of inactivity set by the Administrator. Save edits regularly in the Review Workspace and always select **Sign Out** when leaving the workstation.

4. Uploading Audit Reports

4.1 Before You Upload

Confirm the following to avoid rejection at the validation stage:

- The file is in an accepted format (Section 2.3), is not password-protected and opens normally.
- The report, its annexures, screenshots and logs are collected so that they can be uploaded together as one submission.
- You know the audit type, audited organisation, auditing agency, audit period and submission date.
- The report is the final submitted version, not a working draft.

4.2 Upload Procedure

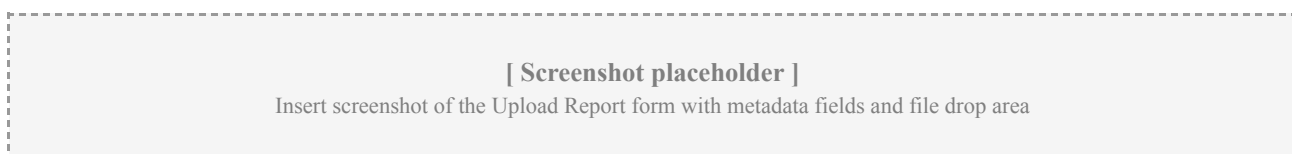


Figure 3: Upload Report screen (indicative)

1. Select **Upload Report** from the menu.
2. Select the **Audit Type**: VAPT, Network Infrastructure, Mobile Application Security, or Application / Web Application Security Testing.
3. Complete the submission details listed in Table 4.3.
4. Select **Choose Files** (or drag files into the drop area) and add the main report first, then annexures.

5. Select **Submit**. The Platform confirms receipt and shows the **Submission ID** and **SHA-256 file hash** for each file.
6. Note the Submission ID; it identifies the report in every later screen and in the audit log.

4.3 Submission Details (Input Requirements)

Field	Required	Description / example
Audit Type	Yes	Selected from the controlled list of four domains.
Auditing Agency	Yes	Empanelled agency name as registered in the Platform.
Audited Organisation	Yes	Name of the organisation whose systems were audited.
Audit Period	Yes	Start and end date of the audit (DD-MM-YYYY).
Report Date	Yes	Date on the report; used for timeliness checks.
Report Version	Yes	e.g. v1.0 or v2.1 (revised after correction).
Audit Context	Optional	Black-box, grey-box or white-box, if known.
Remarks	Optional	Context useful to the reviewer, e.g. "Revised report after agency corrections". Do not enter passwords or sensitive data.

4.4 Security Checks After Upload

Every file is checked before processing. You cannot skip or shorten these checks.

Check	What happens	Possible outcome
File validation	Type, size, integrity and readability are verified.	Accepted, or Rejected with reason.
File hashing	A cryptographic hash is recorded for chain of custody and duplicate detection.	Hash shown; duplicate warning if it already exists.
Malware scan	File is scanned by the approved offline scanning workflow.	Clean, or moved to Quarantine .
Quarantine	Suspect files are isolated and cannot be opened or processed.	Released or deleted only by authorised security staff.

IMPORTANT: Do not re-upload a quarantined file. Report it to your Administrator, quoting the Submission ID.

5. Processing and Tracking Status

5.1 Report Status Lifecycle

Open **Processing Queue** to see all reports available to you. The **Status** column shows where each report is in the pipeline.

Status	Meaning	User action
Uploaded / Validating	Files received; checks in progress.	Wait.
Rejected	A file failed validation.	Read the reason, correct and re-upload.

Status	Meaning	User action
Quarantined	Malware scan raised an alert.	Contact Administrator.
Queued / Processing	OCR, extraction, parsing and mapping running.	Wait; large files take longer.
Ready for Review	Structured output generated.	Reviewer opens the report.
Under Review	A reviewer is verifying the output.	None.
Returned for Correction	Reviewer sent it back.	Analyst corrects submission details or re-uploads.
Approved	Reviewer approved the standard report.	Export if permitted.
Processing Failed	A stage could not complete.	See Section 12.

5.2 Viewing Progress and Actions

- Select a Submission ID to see stage-wise progress, timestamps and any warnings.
- Use the **Filter** bar to narrow the list by audit type, agency, status or date.
- Select **Re-process** to run a failed or updated report again; the earlier run stays in the history.
- Select **Withdraw** to cancel a submission made in error. The action requires a reason and is logged.

6. Reviewing the Structured Output

6.1 Review Workspace Layout

The **Review Workspace** shows the original report on one side and the structured output on the other, so each AI-generated value can be checked against its source.

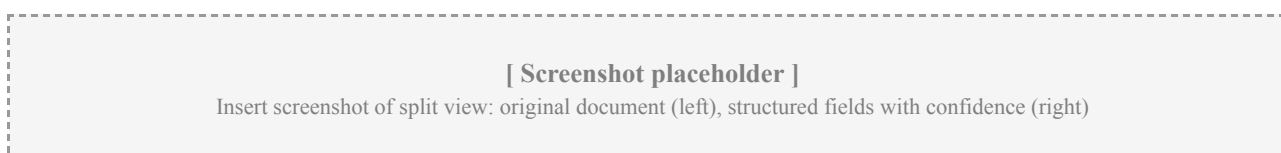


Figure 4: Review Workspace with source view and extracted fields (indicative)

6.2 Source Traceability

Every extracted field carries a **source reference**: source file, page number, section, paragraph, table, screenshot, annexure or log reference. To verify a field:

1. Select the field in the structured panel.
2. The source panel scrolls to the exact location and highlights the text, table cell or image.
3. Compare the value; correct it if it differs.

6.3 Confidence Indicators

Indicator	Meaning	Reviewer attention
High	Rule match or strong model agreement.	Spot-check.
Medium	Model is reasonably sure; wording varied.	Verify against source.
Low	Weak match, poor scan or ambiguous text.	Verify every value; correct if necessary.
Conflict	Rules and AI gave different values.	Choose the correct value; record reason.

NOTE: Confidence levels are guides, not guarantees. A High confidence value can still be wrong; reviewers remain responsible for verification.

6.4 Reviewing Section Mapping

Agencies use different headings for similar content. The **Section Mapping** tab lists each detected heading in the original report and the standard section it was mapped to. Use **Remap** to select a different standard section if the mapping is wrong. Unmapped content is shown under **Unclassified** so that nothing is silently lost.

6.5 Findings and Evidence Linking

The **Findings** tab lists each vulnerability finding with its name, description, severity, affected asset, application and endpoint, CVE / CWE / CVSS details (where available), recommendation and compliance reference. Beneath each finding, **Linked Evidence** shows the screenshots, logs, tool outputs, tables and annexures associated with it.

- Select **Link Evidence** to attach a missed screenshot, log or table to a finding.
- Select **Unlink** to remove an incorrect association.
- A finding without evidence is flagged **No Evidence Linked**; this is recorded and not hidden.

6.6 Approve, Edit, Reject or Override

1. Check the metadata, section mapping, findings and evidence links.
2. To change a value, select the field, choose **Edit**, enter the corrected value and select **Save**. A reason is mandatory for every change.
3. To dismiss an AI output that cannot be relied upon, select **Reject** and give a reason.
4. To replace an AI decision with your own judgement, select **Override** and give the reason.
5. When the review is complete, select **Approve Report** or **Return for Correction**.

Each action is stored in the **Correction History** with user, date and time, old value, new value and reason. Original AI output is preserved and never overwritten.

6.7 Approval Status

Approval status	Description
Draft	Structured output generated; not yet reviewed.
Under Review	Opened by a reviewer.
Returned	Sent back to the Analyst with comments.
Approved	Verified by a reviewer; available to Decision-makers and for export.
Rejected	Output not accepted; report requires re-upload or re-processing.

7. Standard Audit Report Template and Schema

7.1 Template Structure

The standard template defines the section order, mandatory and optional fields, controlled vocabularies, evidence requirements and field descriptions. It serves both human auditors preparing new reports and the Platform's automated processing. Download it from **Standard Template**.

#	Standard section	Status
1	Report metadata (ID, version, dates)	Mandatory
2	Audit type; audited organisation; auditing agency details	Mandatory
3	Audit scope, objectives, assets covered, exclusions and limitations	Mandatory
4	Audit methodology and tools used	Mandatory
5	Executive summary and technical summary	Mandatory
6	Vulnerability findings: affected systems, severity, CVE / CWE / CVSS	Mandatory (CVE/CWE/CVSS where available)
7	Evidence: screenshots, logs, tool outputs	Mandatory for each finding
8	Recommendations and remediation priority	Mandatory
9	Compliance or control mapping	Optional
10	Conclusion	Mandatory
11	Reviewer comments, correction history, approval status	System-maintained

7.2 Domain-Specific Fields

Audit domain	Examples of additional fields
VAPT	Target IPs / URLs, testing phase, exploitation status, proof-of-concept details, tool outputs.
Network Infrastructure Audit	Device inventory, configuration review items, firewall / routing observations, network segments.

Audit domain	Examples of additional fields
Mobile Application Security	Platform (Android / iOS), app version, binary / API analysis, data storage and transport findings.
Application / Web Application Security	Application URL, endpoints, test environment, authentication and session findings, tested vulnerability classes.

The above are indicative; the field guide inside the Platform lists the final fields for each domain.

7.3 Controlled Vocabularies

To keep terms consistent, certain fields accept only listed values. For example, **Severity** is standardised to Critical, High, Medium, Low and Informational. A report that uses “Severe”, “Major” or “Moderate” is mapped to the nearest standard value, and the original label is kept for reference.

7.4 Machine-Readable Schema

The structured report is stored and exported in JSON or XML following the standard schema, which forms the data layer for the other modules. Each element includes traceability fields. The schema supports future extension without disturbing the deployed system (Section 10). A shortened example is given in Appendix B.

8. Rule-Based Parsing and AI-Assisted Extraction

Two complementary components work on each report. Understanding them helps you judge when a value needs closer checking.

8.1 Rule-Based Parsing

Rules identify predictable patterns with high precision:

Rule target	Examples recognised
Dates	12-03-2026, 12 March 2026, 2026-03-12
IP addresses and URLs	192.168.10.5, https://example.gov.in/login
Vulnerability identifiers	CVE-2024-12345, CWE-79
CVSS scores and vectors	7.5, CVSS:3.1/AV:N/AC:L/...
Severity labels	Critical, High, Medium, Low, Informational and common variants
Standard headings and tables	Scope, Methodology, Findings; tabular findings lists

8.2 AI/ML Synonym Mapping and Entity Extraction

AI models handle variation in wording and layout. They map differing headings to a common standard section and extract key entities.

Heading found in report	Mapped to standard section
Findings, Observations, Vulnerabilities, Issues Identified, Security Gaps	Vulnerability Findings
Approach, Testing Approach, Procedure	Audit Methodology

Heading found in report	Mapped to standard section
Summary for Management, Management Overview	Executive Summary
Suggestions, Mitigation, Remedy	Recommendations

Extracted entities include vulnerability name and description, severity, affected asset / application / endpoint, CVE / CWE / CVSS details, evidence, recommendation, compliance reference, conclusion and reviewer-relevant observations.

8.3 How Rules and AI Work Together

- Rules run first; their results are shown with the rule-based tag.
- AI fills gaps and resolves variations; AI results are tagged accordingly with a confidence level.
- If both disagree, the field is marked **Conflict** and left for the reviewer.
- Documentation on rule derivation and rule–AI integration is available through the Help menu.

9. Exporting Outputs

Approved reports can be exported by users with export permission. Select **Exports**, choose a report, choose a format and select **Generate**.

CAUTION: Exports contain sensitive audit data. Save them only on approved storage within the secure environment. Every export is recorded in the audit log, and raw or identifiable audit data must not leave the approved environment.

10. Report-Structure Update Module (Administrators)

When the standard report format changes, the update is handled by a separate decoupled module so the main deployed system is not disturbed. Updates are versioned, reviewed and approved before introduction into the air-gapped system.

1. Prepare the new template / schema version outside the production system and assign a new version number.
2. Submit it for expert review and obtain approval from authorised officials.
3. Package the approved update, generate its checksum and run the malware scan.
4. Transfer the package through the authorised chain-of-custody process.
5. In **Administration > Structure Updates**, import the package, verify the checksum and select **Validate**.
6. Select **Activate** at the approved effective date. The previous version remains available.

NOTE: Reports already processed stay linked to the schema version used at the time. Activating a new version does not change them silently.

11. Administration Basics

Task	Where	Summary
Manage users and roles	Administration > Users	Create, disable and assign roles; register MFA.
View audit logs	Administration > Audit Logs	Uploads, parsing, extraction, review actions, exports and admin changes.
Backup and restore	Administration > Backup	Run or schedule backups and restore from approved backup sets.
Set limits	Administration > Settings	File size, session timeout, accepted formats.

12. Troubleshooting

Try the suggested action first. If the problem persists, contact the support desk (Section 13) with the details listed there.

Problem	Likely cause	Action
Cannot sign in	Wrong credentials or locked account.	Check username and password; if locked, contact Administrator.
MFA code rejected	Code expired or device time is wrong.	Wait for a new code; correct device time; ask Administrator to reset MFA.
Session ends unexpectedly	Inactivity timeout.	Sign in again; save edits more often.
File rejected at validation	Unsupported type, corrupt, protected or too large.	Read the rejection reason; fix the file and re-upload.
File in Quarantine	Malware scan alert.	Do not re-upload; inform Administrator with the Submission ID.
Duplicate file warning	Same hash already uploaded.	Confirm whether this is a new version; do not upload twice.
Report stays in Processing	Large file or busy queue.	Wait; if it remains beyond the expected time, raise a support request.
Processing Failed	A stage could not complete.	Open the Submission ID for the failed stage; select Re-process.
Poor OCR text	Low-resolution or skewed scan.	Obtain a clearer copy and upload a new version.
Table extracted incorrectly	Merged cells or table as image.	Correct the values in Review Workspace using Edit.
Wrong section mapping	Unusual heading wording.	Use Remap and record the reason.
Many Low-confidence fields	Poor document quality or unusual layout.	Review all fields carefully; consider returning the report.
Finding has no evidence	Evidence not detected or not provided.	Use Link Evidence; otherwise leave flagged.

Problem	Likely cause	Action
Export disabled	Report not approved or no permission.	Check approval status and role.
Cannot see a report	Report not assigned to you.	Ask the Administrator for access.

13. Support, Security Practices and FAQs

13.1 Getting Support

Contact the project support desk through the channel specified in the deployment guide. Provide your name and role, Submission ID, the screen where the problem occurred, the exact message shown and the time. **Do not include audit content or credentials in a support request.**

13.2 Security Good Practices

- Use only approved workstations; do not copy report data to personal or removable media.
- Do not upload reports that you have not been authorised to process.
- Do not attempt to bypass validation or malware scanning.
- Review AI output critically; record clear reasons for each correction.
- Sign out when leaving your seat.

13.3 Frequently Asked Questions

Question	Answer
Does the Platform audit a report's quality?	Not in this module. Objective 1 standardises the format; benchmark assessment is covered in a separate manual.
Can the Platform work without the internet?	Yes. It is designed to operate fully offline in an air-gapped environment.
Is AI output final?	No. A reviewer must verify and approve it.
Is the original report changed?	No. The original file is preserved with its hash; the structured output is separate.
What if my report format is new?	Upload as usual. Unrecognised content appears as Unclassified for reviewer handling, and recurring patterns may be addressed in a later update.

Appendix A: Glossary

Term	Meaning
Air-gapped	A network with no direct internet connection.
CVE / CWE / CVSS	Common Vulnerabilities and Exposures / Common Weakness Enumeration / Common Vulnerability Scoring System.
Chain of custody	Documented record of who handled a file and when.
OCR	Optical Character Recognition, converting scanned images to text.

Term	Meaning
Schema	Formal definition of fields and structure of the standard report.
Source traceability	Link from every extracted field to its location in the original report.
Synonym mapping	Mapping of different headings or terms to one standard term.
VAPT	Vulnerability Assessment and Penetration Testing.

Appendix B: Sample Structured Output (Excerpt)

```
{ "report_id": "SUB-000123", "schema_version": "1.0",
  "audit_type": "Web Application Security Testing",
  "findings": [ {
    "finding_id": "F-001",
    "name": "SQL Injection in login form",
    "severity": "High", "severity_original_label": "Major",
    "cwe": "CWE-89", "cvss": 8.1,
    "affected_endpoint": "https://example.gov.in/login",
    "evidence": [ { "type": "screenshot", "ref": "Annexure A, Fig 3" } ],
    "source": { "file": "report.pdf", "page": 14, "section": "4.2" },
    "confidence": "High", "review_status": "Approved"
  } ] }
```

Appendix C: Quick Reference

I want to...	Go to / do this
Upload a report	Upload Report > select audit type > complete details > add files > Submit.
Check status	Processing Queue > select Submission ID.
Verify a value	Review Workspace > select field > compare with highlighted source.
Correct a value	Select field > Edit > enter value and reason > Save.
Approve a report	Review Workspace > Approve Report.
Get the template	Standard Template > Download.
Export	Exports > select report and format > Generate.